ORIE 5630 - Final Project

Joint work by Michael Wachsman (mbw94), Klaus Stier (ks2473), Kexin Deng (kd537) and Olivia Guo (qg77)

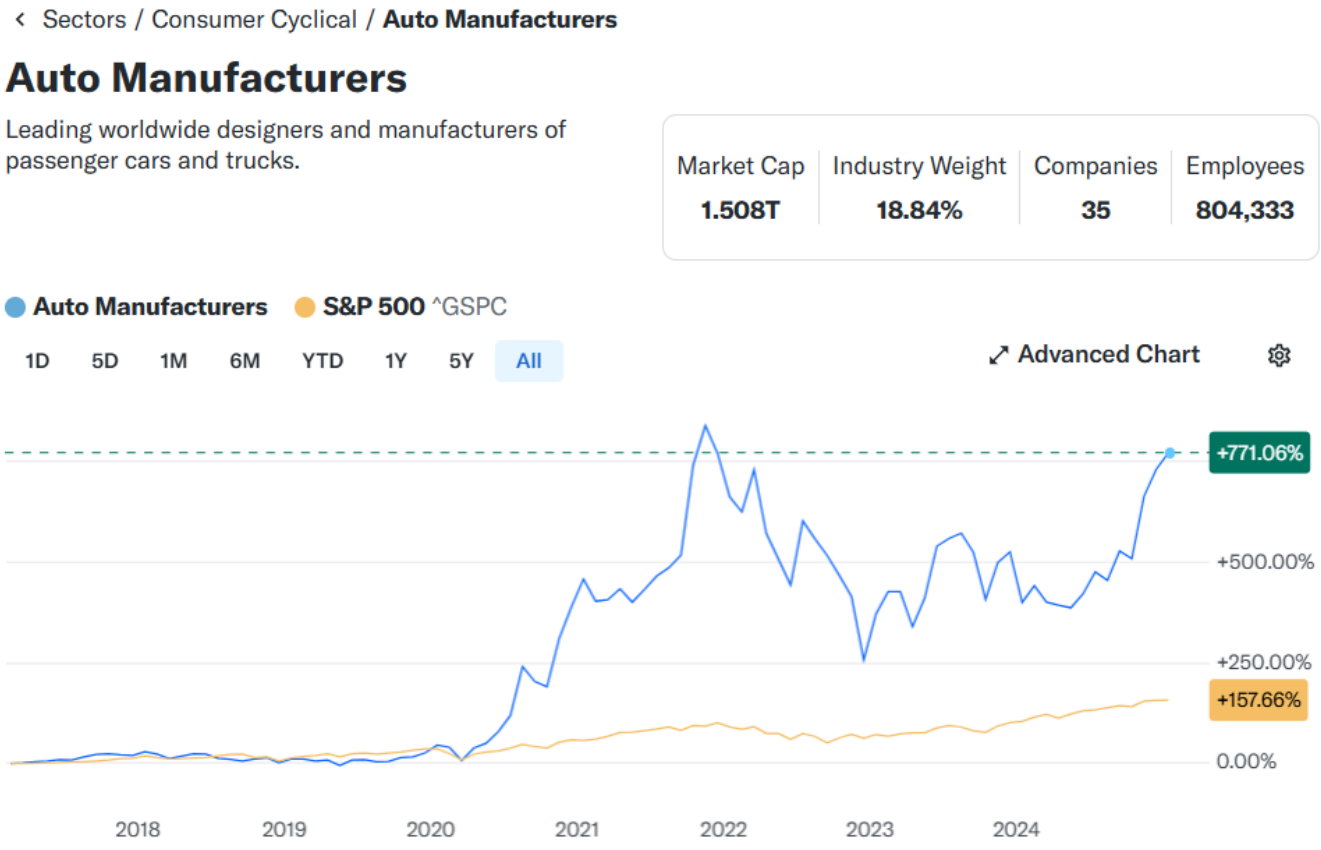
2024-12-13

Analysis of the sector

Consumers Cyclical - Auto Manufacture

Question 1

We start our analysis by considering the industry level data obtained from Yahoo Finance (YH10200020). The index consists of all US auto manufacturing companies listed in the S&P 500, with each company being weighted according to their market cap. The weights are being updated on a regular basis. In particular, this shows that the current weights do not reflect historic weights.



YH10200020 peformance

It can clearly be seen that Tesla (currently) dominates the US auto manufacturing industry.[graphic] As of now, more than 90 % of the weight in the index is put on Tesla stocks. Tesla is (by far) the world's most valuable car manufacturer. Therefore, we choose Tesla to be part of the car manufacturers we wanted to examine closer.

Name	Last Price	1Y Target Est.	Market Weight	Market Cap	Day Change %	YTD Return
TSLA Tesla, Inc.	386.91	222.01	90.62%	1.242T	-0.59%	+55.71%
GM General Motors Co...	53.01	59.28	4.25%	58.291B	-0.75%	+47.58%
F Ford Motor Company	10.61	11.79	3.08%	42.181B	+0.98%	-12.93%
RIVN Rivian Automotive, I...	14.54	15.23	1.08%	14.846B	+11.89%	-38.00%
VFS VinFast Auto Ltd.	5.03	8.00	0.86%	11.764B	+2.86%	-39.90%
LVWR LiveWire Group, Inc.	5.55	7.25	0.08%	1.128B	+2.78%	-50.93%
CYD China Yuchai Intern...	9.47	14.10	0.03%	355.268M	+2.70%	+13.00%

Consitutent assets of the YH10200020 index and their weights as of 12/09/2024

In the index we are considering, Tesla is followed by General Motors (currently 4.25 % of the index weight) and Ford (currently 3.08 % of the index weight). Evidently, these companies have played an important role in the development of the US car industry and are important players till the current day. For instance, the Ford F-series has been the most sold car model in the US for the past years, highlighting the importance of this company. Due to that, we decided to also examine these companies in greater detail.

Below, we downloaded the data necessary for analyzing the 2017 - 2019 period and plotted the cumulative performance of the overall industry, Tesla, General Motors and Ford.

```
# Loading required packages and data
library(quantmod)
```

```
## Warning: Paket 'quantmod' wurde unter R Version 4.3.3 erstellt
```

```
## Lade nötiges Paket: xts
```

```
## Warning: Paket 'xts' wurde unter R Version 4.3.3 erstellt
```

```
## Lade nötiges Paket: zoo
```

```
## Warning: Paket 'zoo' wurde unter R Version 4.3.3 erstellt
```

```
##
## Attache Paket: 'zoo'
```

```
## Die folgenden Objekte sind maskiert von 'package:base':
##
##     as.Date, as.Date.numeric
```

```
## Lade nötiges Paket: TTR
```

```
## Warning: Paket 'TTR' wurde unter R Version 4.3.3 erstellt
```

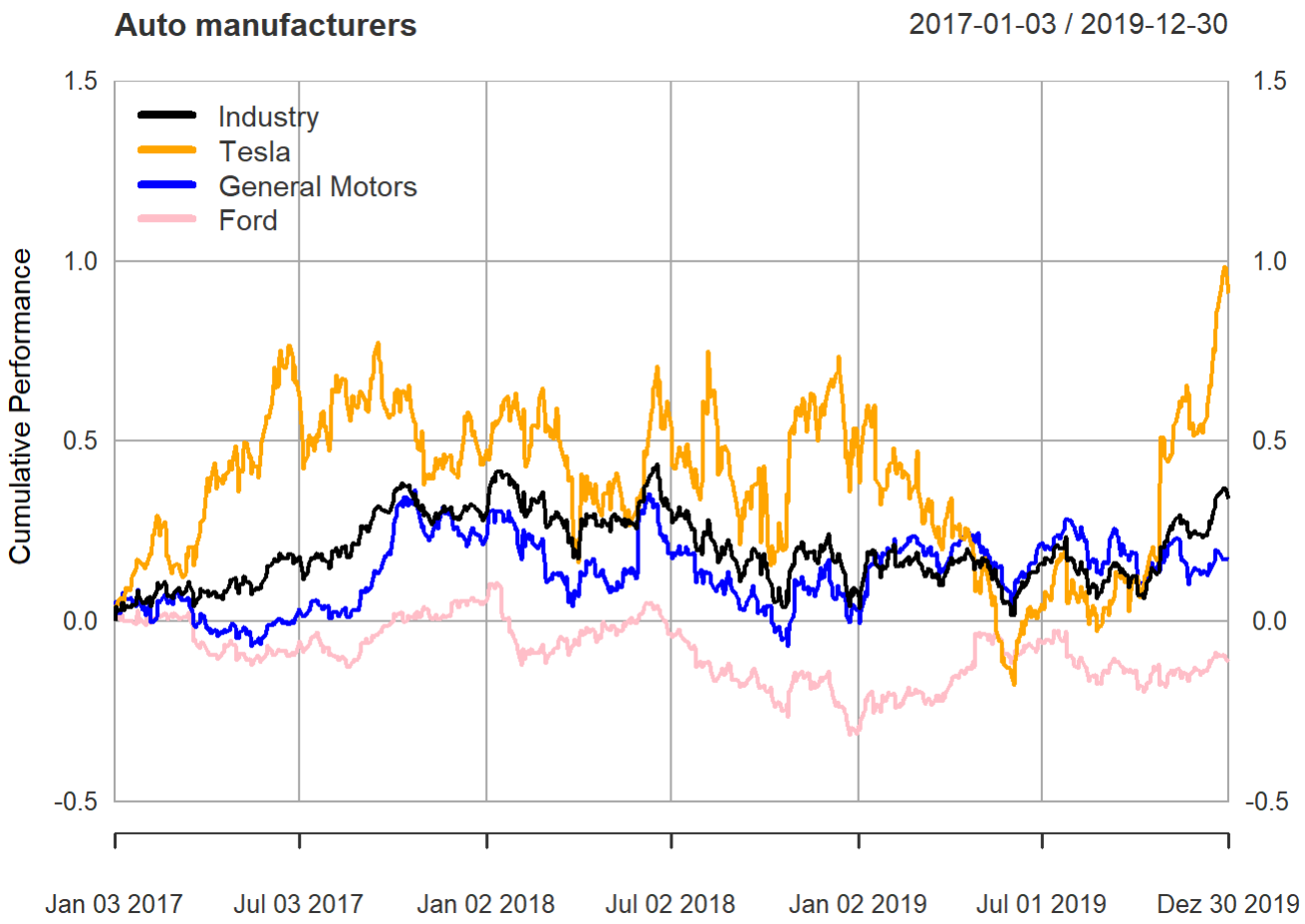
```
## Registered S3 method overwritten by 'quantmod':
##   method            from
##   as.zoo.data.frame zoo
```

```
start_date <- "2017-01-01"
end_date <- "2019-12-31"
industry <- getSymbols("^YH10200020", from = start_date, to = end_date, auto.assign = FALSE)
industry <- industry$YH10200020.Adjusted

in_dai <- dailyReturn(industry, type = "arithmetic")

# Download top 3-4 companies in the industry
# Choice: Tesla, General Motors, Ford
tsla <- getSymbols("TSLA", from = start_date, to = end_date, auto.assign = FALSE)
gm <- getSymbols("GM", from = start_date, to = end_date, auto.assign = FALSE)
ford <- getSymbols("F", from = start_date, to = end_date, auto.assign = FALSE)
tsla <- tsla$TSLA.Adjusted
gm <- gm$GM.Adjusted
ford <- ford$F.Adjusted

ts_dai <- dailyReturn(tsla, type = "arithmetic")
gm_dai <- dailyReturn(gm, type = "arithmetic")
f_dai <- dailyReturn(ford, type = "arithmetic")
head(ts_dai)
```

```
##            daily.returns
## 2017-01-03   0.000000000
## 2017-01-04   0.046085096
## 2017-01-05  -0.001057302
## 2017-01-06   0.009966895
## 2017-01-09   0.009912262
## 2017-01-10  -0.006096497
```

Above, we can see the first entries of the Tesla daily return data. One could similarly print the daily returns for the overall indsutry, Ford or GM.

```
# Compute cumulative returns

in_cum <-  cumprod(in_dai + 1)-1
ts_cum <-  cumprod(ts_dai + 1)-1
gm_cum <-  cumprod(gm_dai + 1)-1
f_cum <-  cumprod(f_dai + 1)-1

pf_start <- 1
in_perf <- pf_start * (1 + in_cum) - 1
ts_perf <- pf_start * (1 + ts_cum) - 1
gm_perf <- pf_start * (1 + gm_cum) - 1
f_perf <- pf_start * (1 + f_cum) - 1

# Plotting cumulative performance

p1 <-  plot.xts(cbind(in_perf, ts_perf, gm_perf, f_perf), type="l",
      col=c("black", "orange","blue","pink"), ylim = c(-0.5, 1.5),
      main="Auto manufacturers", ylab="Cumulative Performance")

addLegend("topleft", c("Industry", "Tesla", "General Motors", "Ford"),
          col=c("black", "orange","blue", "pink"), lty=c(1,1,1,1),lwd=
            c(4, 4, 4, 4))
```



The above graph shows the cumulative performance of the overall industry, Tesla, General Motors and Ford.

We observe that Tesla has outperformed the US auto manufacturing industry as well as Ford and General Motors for almost the entire time period. The value of the Tesla stock has doubled from the beginning of 2017 until the end of 2019, due to a rapid growth in the second half of 2019. The performance of General Motors has

been quite close to the performance of the overall industry. Ford did not perform well during that period, as the cumulative performance of the stock is below 0 for almost the entire period of time, signifying a loss in stock value compared to the beginning of 2017.

Question 2

As a next step, we want to analyze the mean and standard deviation of the daily returns of the overall industry, Tesla, General Motors and Ford in 2017, 2018 and 2019.

```
years <- c("2017", "2018", "2019")

# Reporting mean in one table
in_mean <- sapply(years, \(x){mean(in_dai[x])})
ts_mean <- sapply(years, \(x){mean(ts_dai[x])})
gm_mean <- sapply(years, \(x){mean(gm_dai[x])})
f_mean <- sapply(years, \(x){mean(f_dai[x])})

comb_means <- rbind(in_mean, ts_mean, gm_mean, f_mean)
comb_means <- comb_means * 100 # Turning it into percent values
comb_means_rounded <- round(comb_means, 3)
row.names(comb_means_rounded) <- c("Industry", "TSLA", "GM", "FORD")
print(comb_means_rounded)
```

```
##           2017   2018  2019
## Industry 0.105 -0.051 0.094
## TSLA     0.169  0.093 0.136
## GM       0.086 -0.045 0.062
## FORD     0.025 -0.153 0.116
```

The above table shows the mean returns (for each year) of the daily returns. Notice that the table shows percent values. For instance, we can tell from this table that the mean of the daily returns of Tesla was 0.169 % in 2017.

From the above table, we can see that the average daily returns (in 2018) for GM, Ford and the overall industry have been negative, whereas Tesla exhibits a positive average daily return in 2018.

```
# Reporting sd in one table
in_sd <- sapply(years, \(x){sd(in_dai[x])})
ts_sd <- sapply(years, \(x){sd(ts_dai[x])})
gm_sd <- sapply(years, \(x){sd(gm_dai[x])})
f_sd <- sapply(years, \(x){sd(f_dai[x])})

comb_sd <- rbind(in_sd, ts_sd, gm_sd, f_sd)
comb_sd_rounded <- round(comb_sd, 3)
row.names(comb_sd_rounded) <- c("Industry", "TSLA", "GM", "FORD")
print(comb_sd_rounded)
```

```
##           2017  2018  2019
## Industry 0.011 0.017 0.014
## TSLA     0.022 0.037 0.031
## GM       0.013 0.020 0.015
## FORD     0.011 0.017 0.017
```

The standard deviations in the table above show that 2018 has been the most volatile year. Apart from that, we see that Tesla has the highest standard deviation throughout, which highlights the volatility of this stock.

Question 3

We continue by applying our knowledge from portfolio optimization to find an optimal portfolio (based on the data from the 2017 - 2019 period). In this setting, “optimal” means to ensure the largest possible increase in expected returns per extra unit of risk (variance). Hence, we want to optimize our choice of the weights with respect to the Sharpe ratio and compute the so-called “tangency portfolio”.

The weights of the tangency portfolio can be found using the following formula:

$$w = \frac{\Sigma^{-1}(\mu - r_f \mathbf{1})}{\mathbf{1}^T \Sigma^{-1}(\mu - r_f \mathbf{1})}$$

For the risk-free rate r_f we will use the (mean of) 1-month US treasury bills.

```
# Remark: We want to compute the tangency portfolio in this task

# Preparing data
names(ts_dai) <- "TSLA"
names(gm_dai) <- "GM"
names(f_dai) <- "FORD"
ind_comp <- cbind(ts_dai, gm_dai, f_dai)

# Estimate covariance matrix and mean
sigma.mat <- cov(ind_comp)
mu.vec <- apply(ind_comp, 2, mean)

# Risk-free rate -> Use US-treasury bills
tbill1mo <- getSymbols("DGS1MO", from = start_date, to = end_date,
                      src= "FRED", auto.assign = FALSE)
tbill1mo <- na.omit(tbill1mo)
rate <- as.numeric(mean(tbill1mo))
rf <- rate/365/100 # Rate is annualized and in percent

# Compute optimal portfolio weights
num <-  solve(sigma.mat)%*%(mu.vec-rf)
den <-  as.numeric(t(rep(1,3))%*%solve(sigma.mat)%*%(mu.vec-rf))
weights <-  num/den
weights <- round(weights, 3)
print(weights)
```

```
##        [,1]
## TSLA  1.235
## GM    1.552
## FORD -1.787
```

The computation shows that we will heavily short Ford, which might be due to its underperformance, which we were able to see in the cumulative returns plot for 2017-2019. Moreover, we might infer that Tesla is bought in this portfolio for its overperformance. GM is included due to its comparably good performance (the cumulative performance of GM is above 0 for almost the entire time period considered) and for simultaneously providing some stability in our portfolio.

We continue by analyzing the standard deviation (as a measure of riskiness) as well as the Sharpe ration of our portfolio, the constituent assets and the overall auto industry (we use the abbreviation “AM” to refer to the automotive industry”).

```
# Riskiness - sd
sd_pf <- sqrt(t(weights)%*%sigma.mat%*%weights)
sd_in <- sd(in_dai)
sd_ts <- sd(ts_dai)
sd_gm <- sd(gm_dai)
sd_f <- sd(f_dai)

sds <- c(sd_pf, sd_in, sd_ts, sd_gm, sd_f)
names(sds) <- c("Pf", "AM", "TSLA", "GM", "FORD")
sds_rounded <- round(sds, 3)
print(sds_rounded)
```

```
##    Pf    AM  TSLA    GM  FORD
## 0.045 0.014 0.031 0.017 0.015
```

The above results show that the portfolio we have constructed has a rather high standard deviation, signifying the riskiness connected to this investment. Ford and GM are American auto manufacturers that have been around for more than 100 years, their market value has had a lot of time to stabilize and their products have been fairly consistent for decades. This likely explains why their stock value variances are relatively low.

Tesla is a market shaker who pioneered the electric car industry in the USA. As such, their stock price variance reflects the high highs and low lows associated with innovative companies working in a risky unproven market.

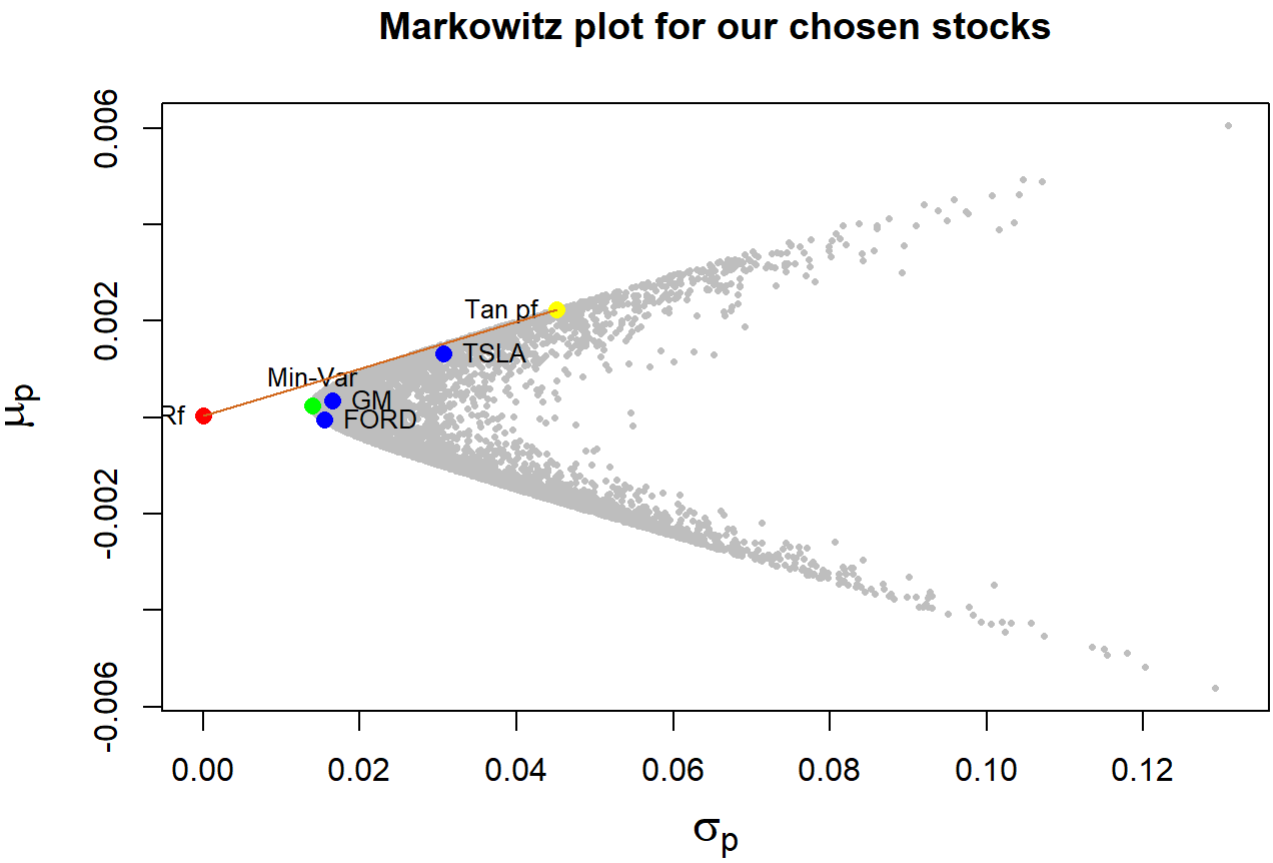
Even though the standard deviation of our optimal portfolio is relatively high compared with each individual asset, the Sharpe ratio of our optimal portfolio makes it worthy, demonstrating the effectiveness of the optimization. The Sharpe ratio values can be found in the table below. Tesla contributes significantly to the portfolio’s returns but also increases its risk.

```
# Riskiness - Sharpe ratio
mean_pf <-  as.numeric(crossprod(weights, mu.vec)) # Return of pf
sharpe_pf <- (mean_pf - rf)/sds[1]
sharpe_in <- (mean(in_dai) - rf)/sds[2]
sharpe_ts <- (mu.vec[1] - rf)/sds[3]
sharpe_gm <- (mu.vec[2] - rf)/sds[4]
sharpe_f <- (mu.vec[3] - rf)/sds[5]

sharpe <- c(sharpe_pf, sharpe_in, sharpe_ts, sharpe_gm, sharpe_f)
names(sharpe) <-  c("Pf", "AM", "TSLA", "GM", "FORD")
sharpe_rounded <- round(sharpe, 3)
print(sharpe_rounded)
```

```
##     Pf     AM   TSLA     GM   FORD
##  0.049  0.032  0.042  0.018 -0.005
```

Using the Markowitz bullet, we can also visualize our findings. The graph below shows the Markowitz bullet, the stocks of Tesla, GM and Ford as well as the optimal portfolio we found (tangency portfolio). Notice that the y-scale, which shows the expected return, is not in per cent this time. The graph supports our finding that the optimal portfolio has a higher risk thna the individual assets, but also a higher expected return.

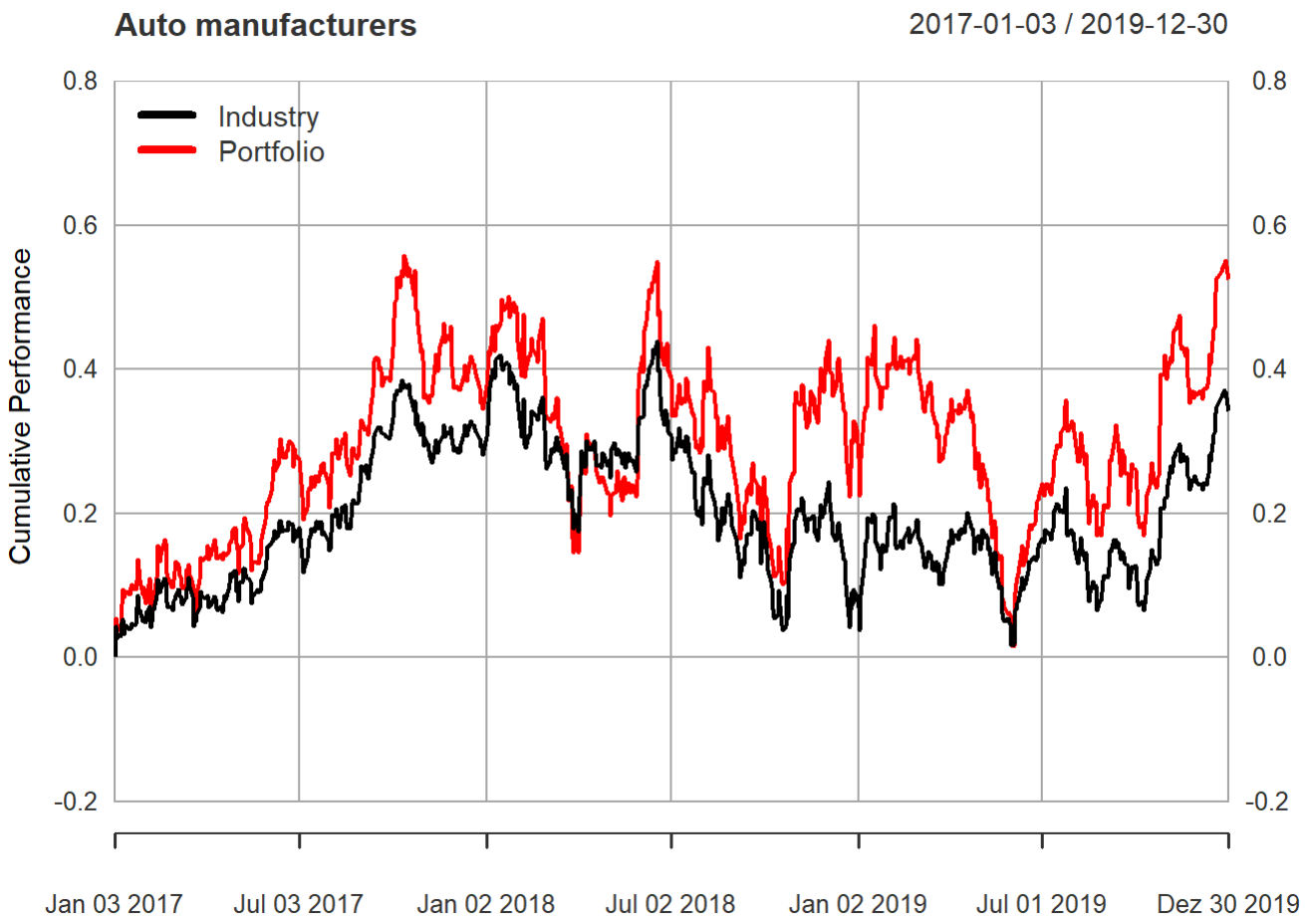
```
n_pf <- 5000
tsla_weight <- rnorm(n_pf)
gm_weight <- rnorm(n_pf)
ford_weight <- 1 - tsla_weight - gm_weight
w <- rbind(tsla_weight, gm_weight, ford_weight)
rownames(w) <- c("TSLA", "GM", "FORD")
sd.vec <- sqrt(diag(sigma.mat))
mu_p <- matrix(NA, n_pf)
sd_p <- matrix(NA, n_pf)

top.mat <- cbind(2*sigma.mat, rep(1,3))
bot.vec <- c(rep(1,3), 0)
Am.mat <- rbind(top.mat, bot.vec)
b.vec <- c(rep(0,3), 1)
zm.mat <- solve(Am.mat)%*%b.vec
wmin.vec <- zm.mat[1:3,1]
mu_gmin <- as.numeric(crossprod(wmin.vec, mu.vec))
sd_gmin <- sqrt(as.numeric(t(wmin.vec)%*%sigma.mat%*%wmin.vec))

for (i in 1:n_pf){
  mu_p[i] <- t(w[,i])%*%mu.vec
  sd_p[i] <- sqrt(t(w[,i])%*%sigma.mat%*%w[,i])
  i = i + 1
}

# plot
ymax <- max(mu_p)
ymin <- min(mu_p)
xmax <- max(sd_p)
plot(sd_p, mu_p, ylim=c(ymin, ymax), xlim=c(0,xmax),
     ylab=expression(mu[p]), xlab=expression(sigma[p]),
     pch=16, col="grey", cex=0.5, cex.lab=1.5)
points(sd.vec, mu.vec, pch=16, cex=1.2, lwd=2, col="blue")
text(sd.vec, mu.vec, labels=c("TSLA", "GM", "FORD"), pos=4, cex=0.8, col="black")
points(sd_gmin, mu_gmin, pch=16, cex=1.2, lwd=2, col="green")
text(sd_gmin, mu_gmin, labels="Min-Var", pos=3, cex=0.8, col="black")
points(0, rf, pch=16, cex=1.2, lwd=2, col="red")
text(0, rf, labels="Rf", pos=2, cex=0.8, col="black")
points(sd_pf, mean_pf, pch=16, cex=1.2, lwd=2, col="yellow")
text(sd_pf, mean_pf, labels="Tan pf", pos=2, cex=0.8, col="black")
lines(c(sd_pf,0), c(mean_pf, rf), lwd=1, col="chocolate")
title("Markowitz plot for our chosen stocks")
```



Question 4

Next, we will have a look at the cumulative performance of our portfolio compared to the overall auto industry for the 2017 - 2019 period.

```
# Daily returns of portfolio found in task 3:
pf_val <- weights[1]*tsla + weights[2]*gm + weights[3]*ford
pf_dai <- dailyReturn(pf_val)
pf_cum <- cumprod(pf_dai + 1) - 1

# Compute performance of portfolio
pf_perf <- pf_start * (1 + pf_cum) - 1

# Plotting - Performance 2017 - 2019
p1 <-  plot.xts(cbind(in_perf, pf_perf), type="l",
                col=c("black","red"), ylim = c(-0.2, 0.8),
                main="Auto manufacturers", ylab="Cumulative Performance")

addLegend("topleft", c("Industry", "Portfolio"),
          col=c("black", "red"), lty=c(1,1),lwd=
            c(4, 4))
```



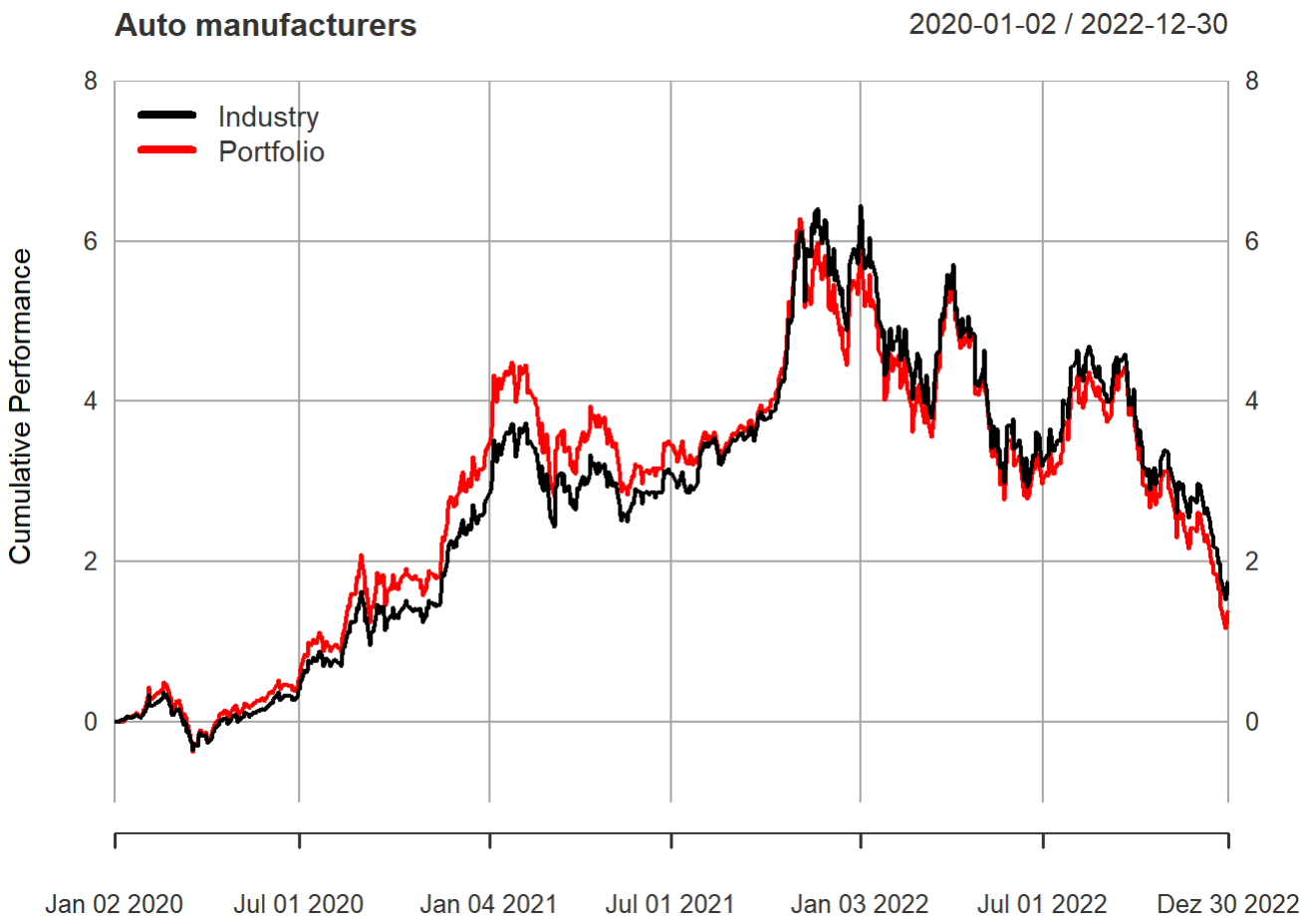
The portfolio outperformed the industry over the 2017-2019 period. The industry showed steady growth but was weighed down by the relatively stagnant performance of traditional automakers like GM and Ford.

Over the 2017-2019 period our portfolio increased in value by 45.7% and outperformed the industry by 8.6%, this could be due to our relatively high weighting of Tesla, given that Tesla's reputation for rapid growth was not fully established until later in the horizon, so their weight in the market portfolio may have reflected their lesser known potential at the time.

The portfolio works well on diversifying risk and capitalizing on Tesla's performance as well as shorting Ford led to higher cumulative returns compared to the industry.

For the 2020 - 2022 period, we observe the following performance:

```
# Comparison: 2020 - 2022
# Need to prepare new data to show performance

start_date_2 <- "2020-01-01"
end_date_2 <- "2022-12-31"
industry_2 <- getSymbols("^YH10200020", from = start_date_2, to = end_date_2, auto.assign = F
ALSE)
industry_2 <- industry_2$YH10200020.Adjusted
in_dai_2 <- dailyReturn(industry_2, type = "arithmetic")

tsla_2 <- getSymbols("TSLA", from = start_date_2, to = end_date_2, auto.assign = FALSE)
gm_2 <- getSymbols("GM", from = start_date_2, to = end_date_2, auto.assign = FALSE)
ford_2 <- getSymbols("F", from = start_date_2, to = end_date_2, auto.assign = FALSE)
tsla_2 <- tsla_2$TSLA.Adjusted
gm_2 <- gm_2$GM.Adjusted
ford_2 <- ford_2$F.Adjusted

ts_dai_2 <- dailyReturn(tsla_2, type = "arithmetic")
gm_dai_2 <- dailyReturn(gm_2, type = "arithmetic")
f_dai_2 <- dailyReturn(ford_2, type = "arithmetic")

# Compute return of our created portfolio
pf_val_2 <- weights[1]*tsla_2 + weights[2]*gm_2 + weights[3]*ford_2
pf_dai_2 <- dailyReturn(pf_val_2)
pf_cum_2 <- cumprod(pf_dai_2 + 1) - 1

# Compute performance of portfolio
pf_perf_2 <- pf_start * (1 + pf_cum_2) - 1

# Compute performance of industry between 2020 - 2022
in_cum_2 <-  cumprod(in_dai_2 + 1)-1
in_perf_2 <- pf_start * (1 + in_cum_2) - 1

# Plotting - Performance 2020 - 2022
p2 <-  plot.xts(cbind(in_perf_2, pf_perf_2), type="l",
                col=c("black","red"), ylim = c(-1, 8),
                main="Auto manufacturers", ylab="Cumulative Performance")

addLegend("topleft", c("Industry", "Portfolio"),
          col=c("black", "red"), lty=c(1,1),lwd=
            c(4, 4))
```



The portfolio slightly outperformed the industry, especially during 2020 and early 2021, when Tesla's value soared. Both the portfolio and industry showed significant volatility and eventual decline at the end of 2022.

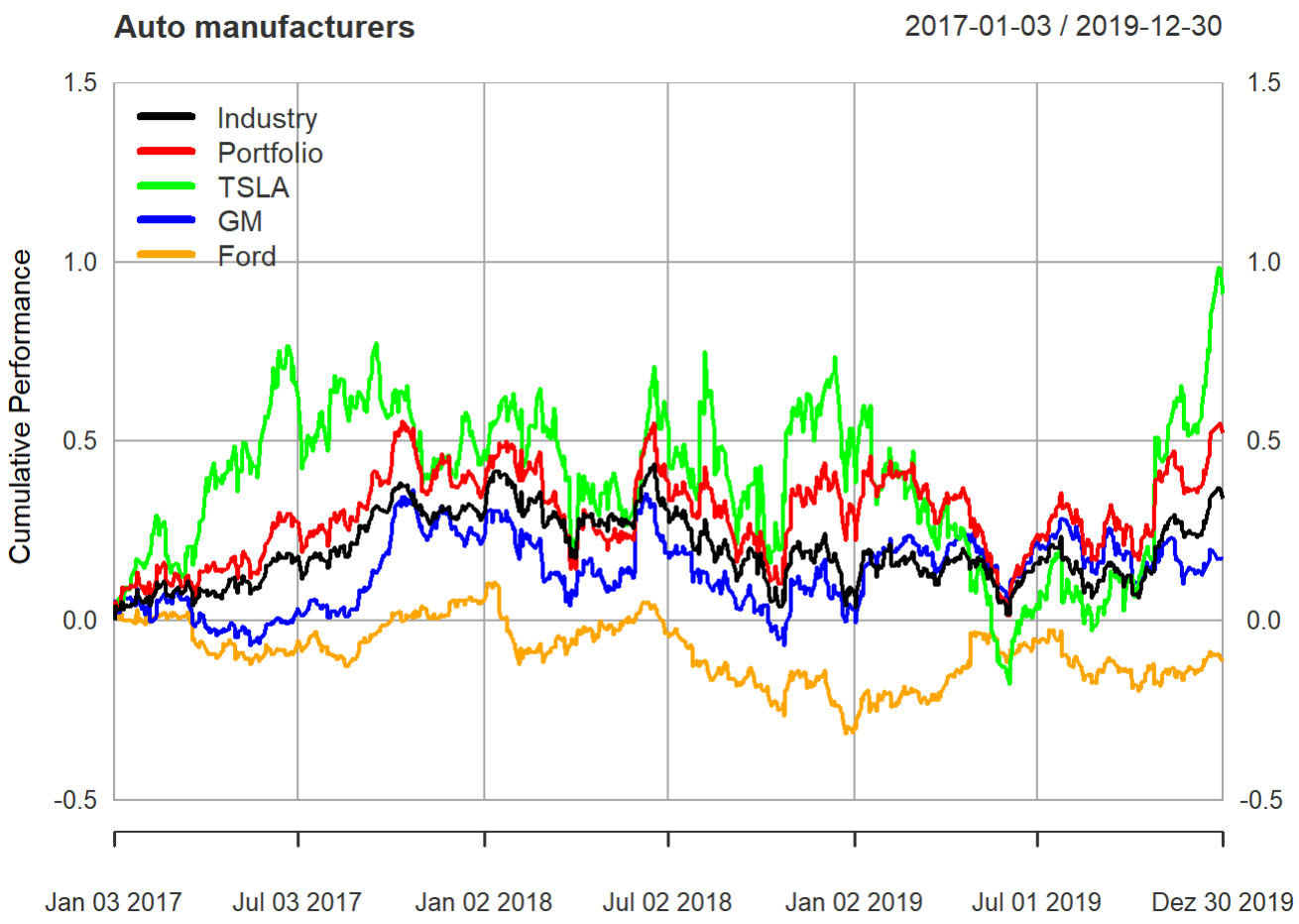
Over the 2020-2022 horizon, our portfolio increased in value by 145%, but it underperformed the market by 11.1%. This may have been due to our shorting of Ford, since the value of Ford stocks started increasing in 2020, eventually reaching a high of approximately 170% of its initial 2020 value in 2022, and finished 2022 up about 25% from 2020. This can be seen in the two graphs on the following pages, which include the performance of Tesla, Ford and GM in addition to the performance of the industry and the portfolio.

This rise was likely due in part to Ford's leadership change in 2020 and the incredible turnout for reservations for the F-150 Lightning, Ford's electric pickup model debuting in 2022. This new leadership was more focused on the production and development of electric vehicles, certainly contributing to the success of the F-150 Lightning. For our portfolio, the good performance of Ford was not beneficial, as we are shorting Ford heavily.

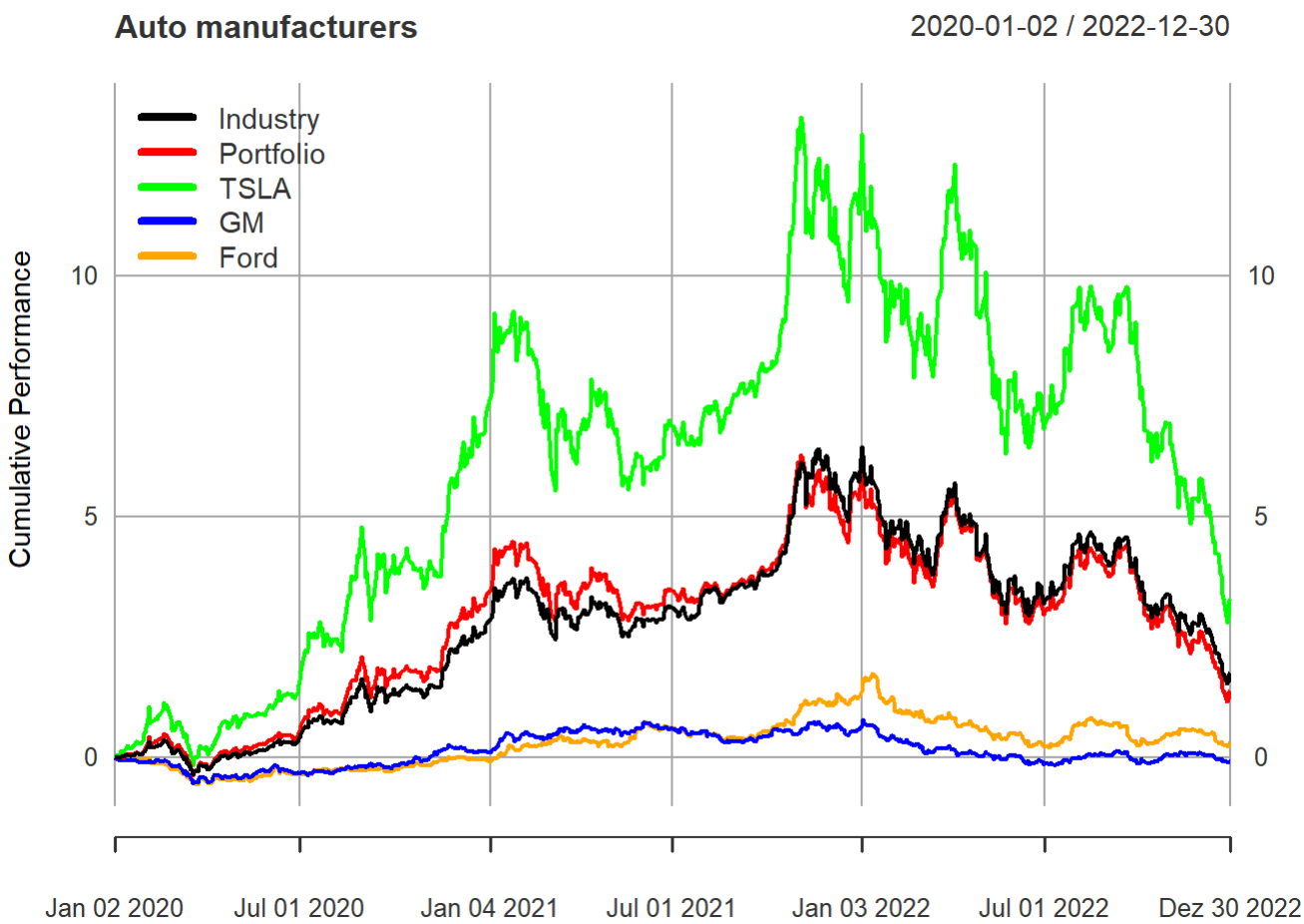
Apart from that, we graphs following below show that in the 2020 - 2022 period, GM was not doing as good as it had previously done. As our portfolio also placed considerable weight on the Ford stock, this might explain why the portfolio was not able to outperform the overall automotive industry in 2020 - 2022.

The below plots are additional work and show the (cumulative) performance of the portfolio and the overall industry together with the performance of Tesla, GM and Ford. These plots are quite insightful, as the allow to connect the performance of the portfolio and the industry to the performance of their constituent assets.

```
# Performance of individual assests, portfolio and industry (2017 - 2019)
ts_cum <-  cumprod(ts_dai + 1)-1
ts_perf <- pf_start * (1 + ts_cum) - 1
gm_cum <-  cumprod(gm_dai + 1)-1
gm_perf <- pf_start * (1 + gm_cum) - 1
f_cum <-  cumprod(f_dai + 1)-1
f_perf <- pf_start * (1 + f_cum) - 1

# Plotting - Performance 2017 - 2019
p3 <-  plot.xts(cbind(in_perf, pf_perf, ts_perf, gm_perf, f_perf), type="l",
                col=c("black","red", "green", "blue", "orange"), ylim = c(-0.5, 1.5),
                main="Auto manufacturers", ylab="Cumulative Performance")

addLegend("topleft", c("Industry", "Portfolio", "TSLA", "GM", "Ford"),
          col=c("black", "red", "green", "blue", "orange"), lty=c(1,1,1,1,1),lwd=
            c(4,4,4,4,4))
```



```
# Performance of individual assets, portfolio and industry (2020 - 2022)
ts_cum_2 <-  cumprod(ts_dai_2 + 1)-1
ts_perf_2 <- pf_start * (1 + ts_cum_2) - 1
gm_cum_2 <-  cumprod(gm_dai_2 + 1)-1
gm_perf_2 <- pf_start * (1 + gm_cum_2) - 1
f_cum_2 <-  cumprod(f_dai_2 + 1)-1
f_perf_2 <- pf_start * (1 + f_cum_2) - 1

# Plotting - Performance 2020 - 2022
p3 <-  plot.xts(cbind(in_perf_2, pf_perf_2, ts_perf_2, gm_perf_2, f_perf_2), type="l",
                col=c("black","red", "green", "blue", "orange"), ylim = c(-1, 14),
                main="Auto manufacturers", ylab="Cumulative Performance")

addLegend("topleft", c("Industry", "Portfolio", "TSLA", "GM", "Ford"),
          col=c("black", "red", "green", "blue", "orange"), lty=c(1,1,1,1,1),lwd=
            c(4,4,4,4,4))
```



One detail that should be noticed is that in the second half of the 2020 - 2022 period, Tesla, the overall industry and our portfolio seem to follow a quite similar pattern of ups and downs.

Question 5

In this part of our analysis, we want to use CAPM to estimate Alphas and Betas for the 2017 - 2019 period.

We recall that the Security Market Line (SCL) is given by

$$R_{i,t} - r_{f,t} = \alpha_i + \beta_i(R_{M,t} - r_{f,t}) + \varepsilon_{i,t}$$

Again, we use 1-month US treasury bills for the risk-free rate. For the market (excess) returns, we are considering the SP500. We fit the daily return data to the above model to obtain Alpha and Beta:

```
# Use CAPM to estimate alphas and betas (2017 - 2019)

# Compute daily risk-free rate based on treasury bills:
rf_dai <- tbill1mo/365/100

# Compute excess returns:
ts_ex <- ts_dai - rf_dai
f_ex <- f_dai - rf_dai
gm_ex <- gm_dai - rf_dai
pf_ex <- pf_dai - rf_dai
in_ex <- in_dai - rf_dai

# Get SP500 data and compute market excess
sp500 <- getSymbols("^GSPC", from = start_date, to = end_date, auto.assign = FALSE)
sp500 <- sp500$GSPC.Adjusted
sp_dai <- dailyReturn(sp500)
sp_ex <- sp_dai - rf_dai

# Compute Alphas and Betas
ts_reg <- lm(ts_ex ~ sp_ex)
ts_alpha <- ts_reg$coefficients[1]
ts_beta <- ts_reg$coefficients[2]

f_reg <- lm(f_ex ~ sp_ex)
f_alpha <- f_reg$coefficients[1]
f_beta <- f_reg$coefficients[2]

gm_reg <- lm(gm_ex ~ sp_ex)
gm_alpha <- gm_reg$coefficients[1]
gm_beta <- gm_reg$coefficients[2]

pf_reg <- lm(pf_ex ~ sp_ex)
pf_alpha <- pf_reg$coefficients[1]
pf_beta <- pf_reg$coefficients[2]

in_reg <- lm(in_ex ~ sp_ex)
in_alpha <- in_reg$coefficients[1]
in_beta <- in_reg$coefficients[2]

# Creating a vector (for overview) and printing
alphas <- c(ts_alpha, f_alpha, gm_alpha, pf_alpha, in_alpha)
names(alphas) <- c("TSLA", "FORD", "GM", "Pf", "AM")
alphas_rounded <- round(alphas, 3)
print(alphas_rounded)
```

```
##   TSLA   FORD     GM     Pf     AM
##  0.001 -0.001  0.000  0.000  0.000
```

The Alphas found above are all quite close to zero. Using the *summary* function, we can extract information about the p-value (for Alpha) from the regression we run. Recall that CAPM states that $\alpha = 0$. The hypothesis we are testing here is:

$H_0 : \alpha = 0$ vs. $H_1 : \alpha \neq 0$

We obtain:

```
pval <- c(summary(ts_reg)$coefficients[7], summary(f_reg)$coefficients[7],
          summary(gm_reg)$coefficients[7], summary(pf_reg)$coefficients[7],
          summary(in_reg)$coefficients[7])
names(pval) <- c("TSLA", "FORD", "GM", "Pf", "AM")
pval_rounded <- round(pval, 3)
print(pval_rounded)
```

```
##  TSLA  FORD    GM    Pf    AM
## 0.488 0.216 0.647 0.806 0.916
```

The p-values above are all larger than 0.05 (considerably). This shows that CAPM indeed holds for all companies during the 2017 - 2019 period.

Apart from that, we see that Tesla has the highest Beta in this period. As Tesla's Beta is larger than 1, Tesla can be considered an aggressive stock. On the contrary, we notice that GM, Ford and the overall automotive industry have a Beta which is quite close to 1. This shows that the excess returns of GM, Ford and the overall industry behave somewhat similar to those of the SP500. Lastly, it should be mentioned that the Beta of Ford is smaller than 1, signifying that Ford is a rather non-aggressive stock in the 2017 - 2019 period. All Betas can be found in the table below:

```
betas <- c(ts_beta, f_beta, gm_beta, pf_beta, in_beta)
names(betas) <- c("TSLA", "FORD", "GM", "Pf", "AM")
betas_rounded <- round(betas, 3)
print(betas_rounded)
```

```
##  TSLA  FORD    GM    Pf    AM
## 1.341 0.953 1.057 1.185 1.039
```

Question 6

Next, we want to use CAPM to estimate Alphas and Betas for the 2020 - 2022 period as well.

```r
# Use CAPM to estimate alphas and betas (2020 - 2022)

# Compute daily risk-free rate based on treasury bills:
tbill1mo_2 <- getSymbols("DGS1MO", from = start_date_2, to = end_date_2,
                        src= "FRED", auto.assign = FALSE)
tbill1mo_2 <- na.omit(tbill1mo_2)
rf_dai_2 <- tbill1mo_2/365/100

# Compute excess returns:
ts_ex_2 <- ts_dai_2 - rf_dai_2
f_ex_2 <- f_dai_2 - rf_dai_2
gm_ex_2 <- gm_dai_2 - rf_dai_2
pf_ex_2 <- pf_dai_2 - rf_dai_2
in_ex_2 <- in_dai_2 - rf_dai_2

# Get SP500 data and compute market excess
sp500_2 <- getSymbols("^GSPC", from = start_date_2, to = end_date_2, auto.assign = FALSE)
sp500_2 <- sp500_2$GSPC.Adjusted
sp_dai_2 <- dailyReturn(sp500_2)
sp_ex_2 <- sp_dai_2 - rf_dai_2

# Compute Alphas and Betas
ts_reg_2 <- lm(ts_ex_2 ~ sp_ex_2)
ts_alpha_2 <- ts_reg_2$coefficients[1]
ts_beta_2 <- ts_reg_2$coefficients[2]

f_reg_2 <- lm(f_ex_2 ~ sp_ex_2)
f_alpha_2 <- f_reg_2$coefficients[1]
f_beta_2 <- f_reg_2$coefficients[2]

gm_reg_2 <- lm(gm_ex_2 ~ sp_ex_2)
gm_alpha_2 <- gm_reg_2$coefficients[1]
gm_beta_2 <- gm_reg_2$coefficients[2]

pf_reg_2 <- lm(pf_ex_2 ~ sp_ex_2)
pf_alpha_2 <- pf_reg_2$coefficients[1]
pf_beta_2 <- pf_reg_2$coefficients[2]

in_reg_2 <- lm(in_ex_2 ~ sp_ex_2)
in_alpha_2 <- in_reg_2$coefficients[1]
in_beta_2 <- in_reg_2$coefficients[2]

# Creating a vector (for overview) and printing
alphas_2 <- c(ts_alpha_2, f_alpha_2, gm_alpha_2, pf_alpha_2, in_alpha_2)
names(alphas_2) <- c("TSLA", "FORD", "GM", "Pf", "AM")
alphas_2_rounded <- round(alphas_2, 3)
print(alphas_2_rounded)
```

```
##  TSLA  FORD    GM    Pf    AM
## 0.002 0.000 0.000 0.001 0.001
```

Again, we find that all Alphas are quite close to zero. Below, we listed the p-values for Alpha. As all p-values are larger than 0.05, this suggests that CAPM holds (i.e. α = 0).

```
pval_2 <- c(summary(ts_reg_2)$coefficients[7], summary(f_reg_2)$coefficients[7],
          summary(gm_reg_2)$coefficients[7], summary(pf_reg_2)$coefficients[7],
          summary(in_reg_2)$coefficients[7])
names(pval_2) <- c("TSLA", "FORD", "GM", "Pf", "AM")
pval_2_rounded <- round(pval_2, 3)
print(pval_2_rounded)
```

```
##  TSLA  FORD    GM    Pf    AM
## 0.089 0.626 0.928 0.226 0.157
```

We continue by examining the Betas. Evidently, all Betas are larger than 1, which shows that Tesla, GM and Ford as well as our portfolio and the overall automotive industry can be considered to be rather aggressive.

```
betas_2 <- c(ts_beta_2, f_beta_2, gm_beta_2, pf_beta_2, in_beta_2)
names(betas_2) <- c("TSLA", "FORD", "GM", "Pf", "AM")
betas_2_rounded <- round(betas_2, 3)
print(betas_2_rounded)
```

```
##  TSLA  FORD    GM    Pf    AM
## 1.455 1.209 1.285 1.458 1.369
```

The below table lists the Alphas and Betas found for both the 2017 - 2019 period as well as the 2020 - 2022 period.

By comparing the Betas of this two time periods, we realize that all Betas have increased in the 2020 - 2022 period (compared to 2017 - 2019). This might be an indicator that the 2020 - 2022 period has been challenging and came with some instability (e.g. COVID19-pandemic). Tesla still has a high Beta in 2020-2022, but also the Beta of our portfolio and the overall automonufacturing industry has increased considerably.

```
# Combining data from both periods
comb_table <- rbind(alphas_rounded, betas_rounded, alphas_2_rounded,  betas_2_rounded)
row.names(comb_table) <- c("Alpha (2017-2019)", "Beta (2017-2019)", "Alpha (2020-2022)", "Bet
a (2020-2022)")
print(comb_table)
```

```
##                     TSLA   FORD    GM    Pf    AM
## Alpha (2017-2019) 0.001 -0.001 0.000 0.000 0.000
## Beta (2017-2019)  1.341  0.953 1.057 1.185 1.039
## Alpha (2020-2022) 0.002  0.000 0.000 0.001 0.001
## Beta (2020-2022)  1.455  1.209 1.285 1.458 1.369
```

Question 7

We continue our analysis by estimating the value at risk (VaR) and the expected shortfall (ES) for the 2017 - 2019 period. We consider a level of significance of 5% and an initial investment of S = $1000. Furthermore, it should be mentioned that we will make use of the non-parametric method in this estimation. Notice that VaR as well as ES are dollar values.

Before we move on to our results, we quickly recall the definition of VaR and ES. We let L signify the loss. For $0 < \alpha < 1$, VaR(α) can be characterized by the following equation:

$$P\big(L > VaR(\alpha)\big) = \alpha$$

The expected shortfall (with respect to the level of significance α) is defined as:

$$ES(\alpha) = E\big[L|L > VaR(\alpha)\big]$$

After this quick recap, we can start to estimate VaR and ES for the 2017 - 2019 period:

```
# Amount invested and level of significance
S <- 1000
alp <- 0.05

# Compute VaR and ES (for 2017 - 2019)
VaR_ts <-  -S*as.numeric(quantile(ts_dai,alp))
L_ts <-  -S*ts_dai
ES_ts <-  mean(L_ts[L_ts>VaR_ts])

VaR_f <-  -S*as.numeric(quantile(f_dai,alp))
L_f <-  -S*f_dai
ES_f <-  mean(L_f[L_f>VaR_f])

VaR_gm <-  -S*as.numeric(quantile(gm_dai,alp))
L_gm <-  -S*gm_dai
ES_gm <-  mean(L_gm[L_gm>VaR_gm])

VaR_in <-  -S*as.numeric(quantile(in_dai,alp))
L_in <-  -S*in_dai
ES_in <-  mean(L_in[L_in>VaR_in])

# Creating a table
VaR_raw <- c(VaR_ts, VaR_f, VaR_gm, VaR_in)
VaR <- sapply(VaR_raw, \(x){round(x, digits=3)})
names(VaR) <- c("TSLA", "FORD", "GM", "AM")

ES_raw <- c(ES_ts, ES_f, ES_gm, ES_in)
ES <- sapply(ES_raw, \(x){round(x, digits=3)})
names(ES) <- c("TSLA", "FORD", "GM", "AM")

VaR_ES_table <- rbind(VaR, ES)
print(VaR_ES_table)
```

```
##       TSLA   FORD     GM     AM
## VaR 43.141 23.477 27.901 22.592
## ES  68.172 37.119 36.621 31.198
```

From the table above, we see that Tesla has (by far) the highest VaR and ES. This again underlines that Tesla stocks (in 2017 - 2019) were considerably riskier than Ford stocks or GM stocks (or the overall auto manufacturing industry).

Question 8

In analogy to question 7, we now want to estimate VaR and ES for the 2020 - 2022 period.

```
# Compute VaR and ES (for 2020 - 2022)
VaR_ts_2 <-  -S*as.numeric(quantile(ts_dai_2,alp))
L_ts_2 <-  -S*ts_dai_2
ES_ts_2 <-  mean(L_ts_2[L_ts_2>VaR_ts_2])

VaR_f_2 <-  -S*as.numeric(quantile(f_dai_2,alp))
L_f_2 <-  -S*f_dai_2
ES_f_2 <-  mean(L_f_2[L_f_2>VaR_f_2])

VaR_gm_2 <-  -S*as.numeric(quantile(gm_dai_2,alp))
L_gm_2 <-  -S*gm_dai_2
ES_gm_2 <-  mean(L_gm_2[L_gm_2>VaR_gm_2])

VaR_in_2 <-  -S*as.numeric(quantile(in_dai_2,alp))
L_in_2 <-  -S*in_dai_2
ES_in_2 <-  mean(L_in_2[L_in_2>VaR_in_2])

# Creating a table
VaR_raw_2 <- c(VaR_ts_2, VaR_f_2, VaR_gm_2, VaR_in_2)
VaR_2 <- sapply(VaR_raw_2, \(x){round(x, digits=3)})
names(VaR_2) <- c("TSLA", "FORD", "GM", "AM")

ES_raw_2 <- c(ES_ts_2, ES_f_2, ES_gm_2, ES_in_2)
ES_2 <- sapply(ES_raw_2, \(x){round(x, digits=3)})
names(ES_2) <- c("TSLA", "FORD", "GM", "AM")

VaR_ES_table_2 <- rbind(VaR_2, ES_2)
row.names(VaR_ES_table_2) <- c("VaR", "ES")
print(VaR_ES_table_2)
```

```
##       TSLA   FORD     GM     AM
## VaR 68.035 43.189 45.696 56.599
## ES  98.761 68.044 66.847 80.320
```

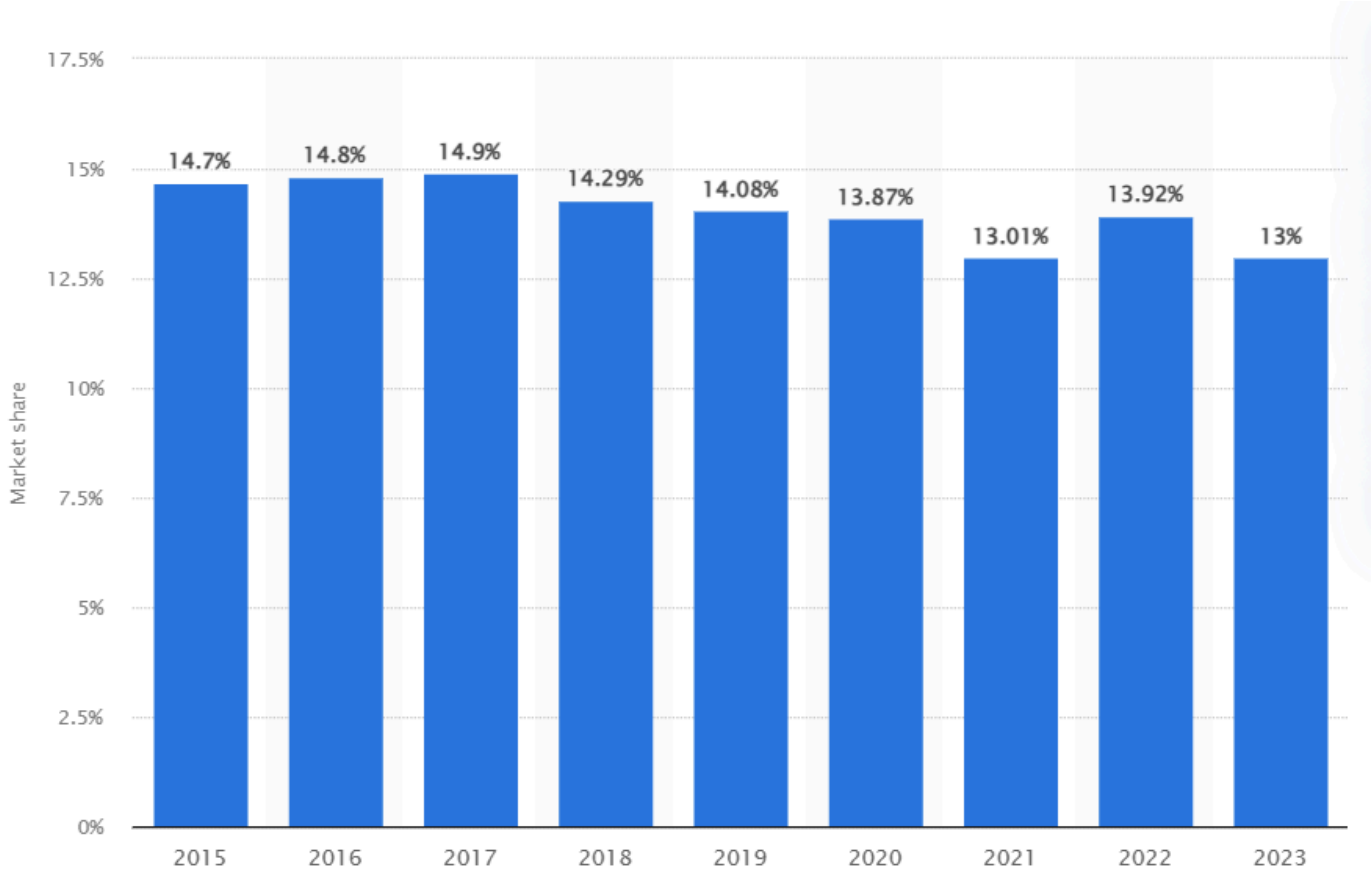
From the above table, we see that Tesla still has the highest VaR and the highest ES. It is noticeable that every VaR and every ES has increased in the 2020 - 2022 period (compared to 2017 - 2019). This underlines that the 2020 - 2022 period was challenging and that risk has increased in this period of time. A possible reason for this might be the COVID19-pandemic.

Apart from that, it is interesting that the VaR and ES of the (overall) automotive industry has increased considerably. In fact this VaR and ES is larger than the VaR and ES of GM or Ford. A possible explanation for this lies in the way the YH10200020 index is constructed. The weighting of these assets is done according to their market cap. As Tesla stock has performed extraordinarily well from January 2020 - January 2022, more and more weight was placed on the (riskier) Tesla stocks, which might have resulted in a higher VaR and ES for the YH10200020 market index.

Question 9

In the portfolio optimization, we sought to maximize the Sharpe ratio, ensuring the largest possible increase in expected returns per extra unit of risk (variance). The portfolio which was found to be optimal based on the 2017 - 2019 data (see Question 3) puts a negative weight on Ford, while buying Tesla and Ford.

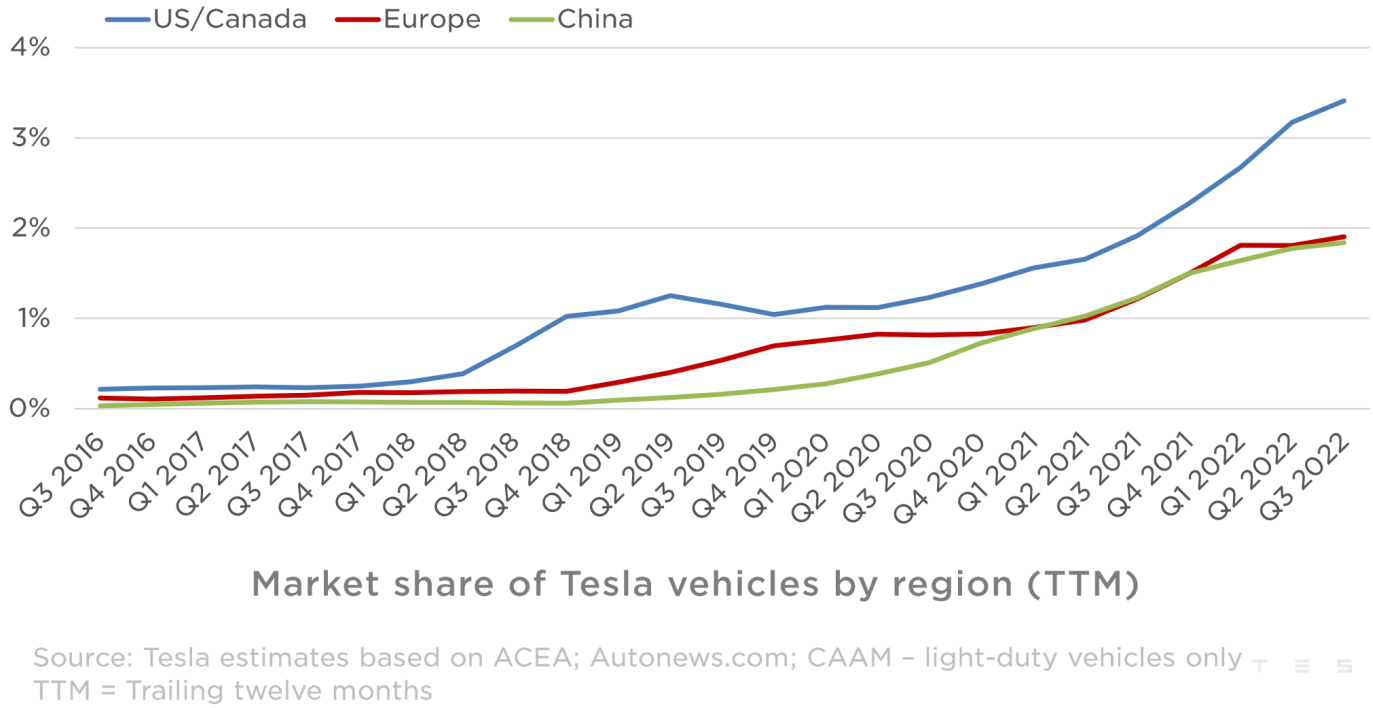
The shorting on Ford makes sense, as the Ford stock was the only one of the three considered stocks to decrease in value over the 2017-2019 horizon. This is likely partially due to Ford's declining sales in 2018. Also, over that three year period Ford's market share slowly declined, which can be seen in this graph:



Market share of Ford in the United States, source: https://www.statista.com/statistics/239614/vehicle-sales-market-share-of-ford-in-the-united-states/ (https://www.statista.com/statistics/239614/vehicle-sales-market-share-of-ford-in-the-united-states/)

While the overall decrease was less than 1%, it is likely that this stagnation may have turned away more aggressive investors and those looking to make gains over shorter horizons. Apart from that, Ford was involved in a Mass Action Lawsuit regarding the powershift transmission which was used in some Ford models. The uncertainty connected to this lawsuit might as well have negatively affected the performance of the Ford stock during the 2017 - 2019 period.

A positive loading was placed on GM, which makes sense given that GM had a positive average return over the 2017-2019 period, with very little variance compared to Tesla, so it may have provided extra stability to our portfolio. Tesla has a positive weight because of its immense growth of returns over the 2017-2019 horizon. They also increased their market share by around 4 times over this period, so this sign of growth likely signaled continued growth for the company, attracting more ambitious investors. The growth of Tesla's market share can be seen in this graph:



Market share of Tesla in different markets, source: https://insideevs.com/news/617556/tesla-market-share-2022q3/ (https://insideevs.com/news/617556/tesla-market-share-2022q3/)

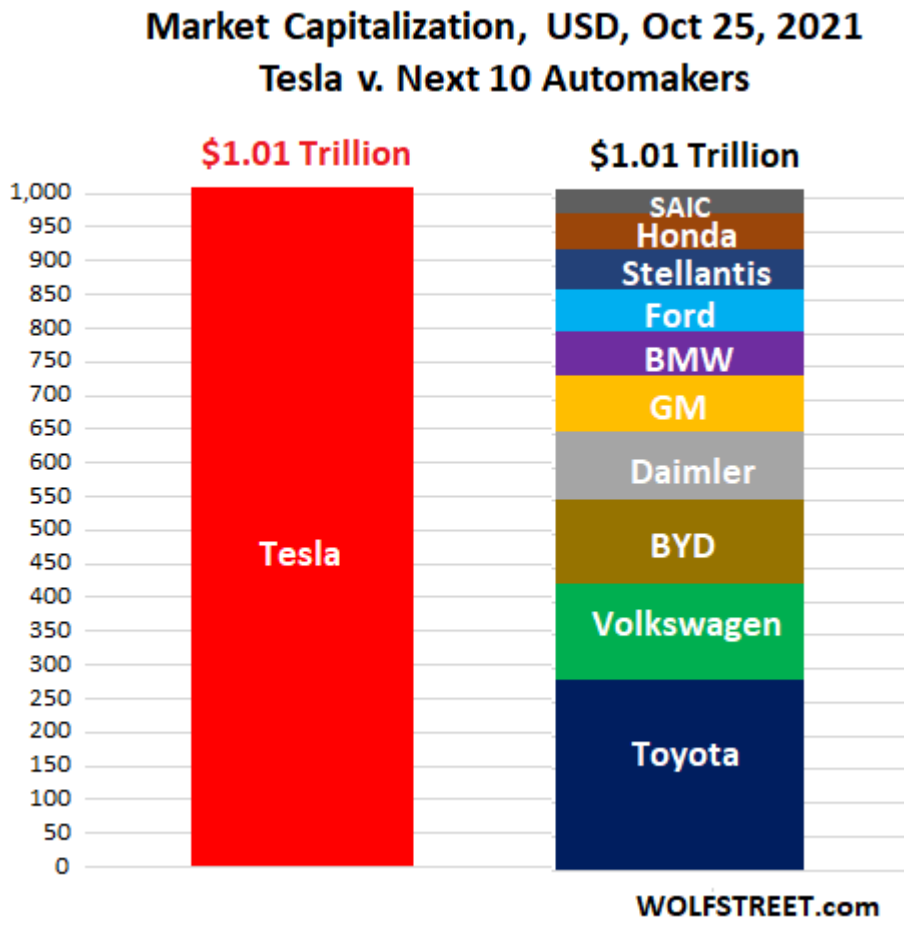
Over the 2017-2019 period our portfolio increased in value by 47.8% and outperformed the industry by 10.1%. Over the 2020-2022 horizon however, our portfolio increased in value by 138%, but it underperformed the market.

This may have been due to our shorting of Ford, since the value of Ford stocks started increasing in 2020, eventually reaching a high of approximately 170% of its initial 2020 value in 2022, and finished 2022 up about 25% from 2020. This rise was likely due in part to Ford's leadership change in 2020 and the incredible turnout for reservations for the F-150 Lightning, Ford's electric pickup model debuting in 2022 (see e.g. https://www.cnbc.com/2022/01/03/ford-beats-tesla-to-become-auto-industrys-top-growth-stock-in-2021.html (https://www.cnbc.com/2022/01/03/ford-beats-tesla-to-become-auto-industrys-top-growth-stock-in-2021.html)). This new leadership of Ford was more focused on the production and development of electric vehicles, certainly contributing to the success of the F-150 Lightning. This might explain some of our portfolio's performance during that time.

Apart from that, an analysis of the Betas (see Question 5 and 6) as well as the tail risks (see Question 7 and 8) highlights that the 2020 - 2022 period has been particularly challenging for the automotive industry. Compared to the 2017 - 2019 period, all Betas, VaRs and expected shortfalls have increased. This underlines that the tail risk in the sector has increased for the 2020 - 2022 period. Partially, this might be due to the COVID19-pandemic. In March 11th 2020, the WHO declared the spread of the Corona virus to be a global pandemic situation. The stock markets crashed at this time and all the stocks considered lost roughly 50% of their value in this crash (see Question 4).

As we have seen in Question 4, the automotive industry recovered from the crash caused by COVID19. It can clearly be seen that Tesla's recovery was faster and the gains of the Tesla stock between April 2020 and November 2021 have been tremendous. A reason for that might be that Tesla has always been a market shaker, trailblazing the way for the electric car industry as a whole. Tesla is also not only considered a car manufacturer. For instance, Tesla is a part of the NASDAQ100, which shows that Tesla is also perceived as a Tech company. This also explains why the Betas and VaRs of Tesla are higher than the VaRs and Betas of Ford and GM. While Ford and GM are well-established players in the car industry, Tesla is a rather new company, which is both disruptive and innovative.

Another observation which can be made is that the Tesla stock was not doing well in the second half of 2022. There is an entire bundle of reasons for this. Among the reasons is for instance the Twitter purchase of Tesla CEO Elon Musk. He sold Tesla stocks worth billions of dollars, which decreased the stock price of Tesla and also disappointed some investors. Apart from that, the lockdown in Shanghai caused the Tesla factory there to shut down, which lowered production. In addition to that, some problems connected to the new Gigafactory in Berlin occurred, which negatively affected the stock price as well.



Tesla's market cap campared to the market cap of other car makers. source: https://wolfstreet.com/2021/10/26/teslas-market-cap-gigantic-v-next-10-automakers-v-teslas-global-market-share-minuscule/ (https://wolfstreet.com/2021/10/26/teslas-market-cap-gigantic-v-next-10-automakers-v-teslas-global-market-share-minuscule/)

Overall, our analysis has shown that the years 2020 - 2022 have been challenging for the US auto manufacturing industry and risks in the industry have increased. Tesla has been pioneering in the field of electric vehicles and therefore their stock price has increased massively, which resulted in Tesla having a very large market cap (see also graph above). With its enormous growth, Tesla has also significantly contributed to the performance of the overall industry, which is captured for instance in the performance of the YH10200020 index.